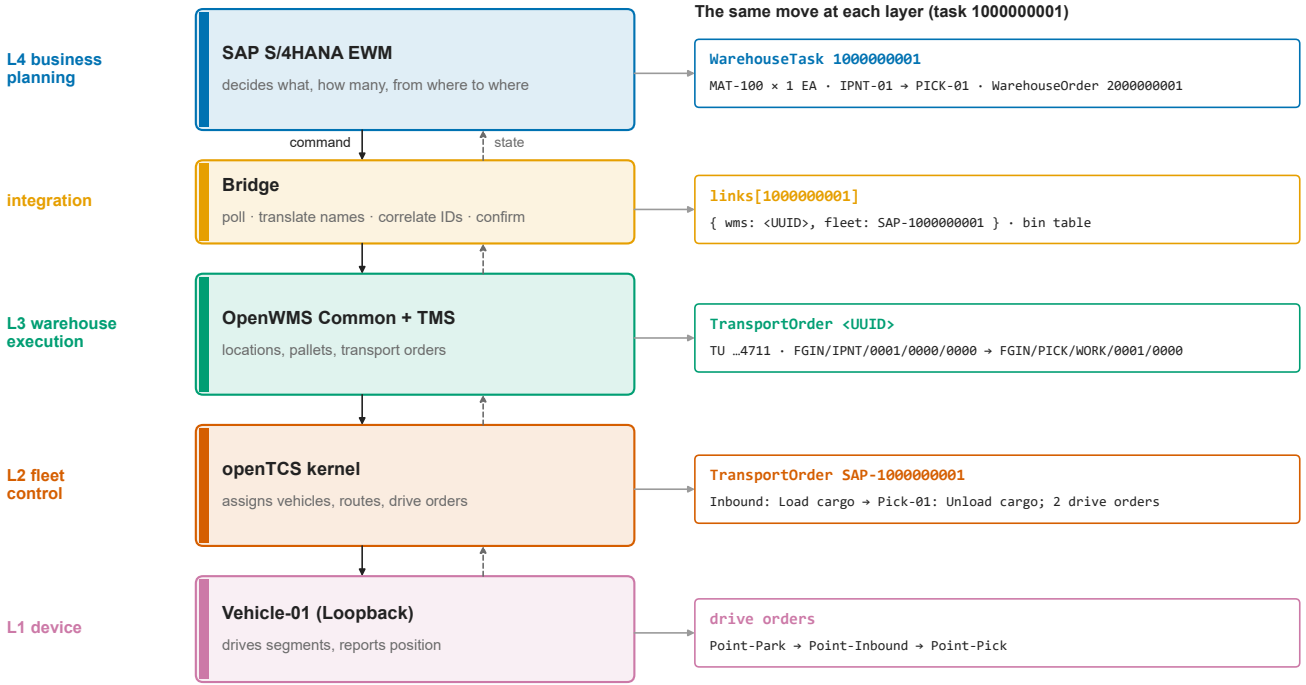
a **Layers: commands are refined downwards, states reported upwards**



Lower layers are more concrete: quantities → pallets/locations → stations → route points.

b **Two mappings kept by the bridge**

Layer	Identifier	Source	Target
SAP	1000000001	IPNT-01	PICK-01
OpenWMS	<UUID, per run>	FGIN/IPNT/0001/…	FGIN/PICK/WORK/…
openTCS	SAP-1000000001	Inbound	Pick-01
Vehicle	Vehicle-01	Point-Inbound	Point-Pick

Stage	SAP	OpenWMS	openTCS	Vehicle
waiting	''	–	–	IDLE
running	''	STARTED	BEING_PROCESSED	EXEC.
done	'C'	FINISHED	FINISHED	IDLE
cancel	'A'	CANCELED	FAILED (withdrawn)	–

An EWM warehouse task is only '', 'C' or 'A'; while the vehicle runs SAP stays open, so progress is visible only downstream. Levels follow ISA-95.